



Artificial Intelligence Foundations (CIS4049-N)

Week 1 – What's AI – Part 2

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What is “*Intelligence*”

- Latin: *intelligere* - understand, comprehend
- Intelligence is a descriptive concept - it describes certain properties of an individual or a group of individuals
- There is no consensus on the definition of intelligence
- Most definitions include concepts such as abstract reasoning, understanding, self-consciousness, communication, learning, planning, and problem solving

“Intelligence” – Some definitions

- “ability to **adapt** oneself adequately to relatively new situations in life.” (R. Pintner)
- “having **learned** or the ability to **learn** to adjust oneself to the environment.” (Colvin)
- “the ability to carry out **abstract thinking**.” (Terman)
- “innate general **cognitive ability**” (Burt)
- “**appropriate** and **adaptable behavior** in given circumstances.” (Psihologija, group of authors, SK, Zagreb, 1992)
- “it manifests itself only relative to specific social and cultural contexts.” (J. Weizenbaum, 1975)

A pragmatic approach to intelligence

- If a given kind of behavior (by a human, ant, elephant, robot, etc.) is interesting – *how did it come to being?*
- Such an approach enables us to understand the fundamental principles of artificial intelligence: “Understanding by building”
- **Cognitive science** - an interdisciplinary study of the mind

What is Artificial Intelligence

- Branch of **Computer Science**
- The branches of Artificial Intelligence (according to Association of Computing Machinery, ACM):
 - (1) General AI (cognitive modeling, philosophical foundations)
 - (2) Expert systems and applications
 - (3) Automated programming
 - (4) Deduction and theorem proving
 - (5) Formalisms and methods for knowledge representation
 - (6) Machine learning
 - (7) Understanding and processing of natural and artificial languages
 - (8) Problem solving, control methods, and state space search
 - (9) Robotics
 - (10) Computer vision, pattern recognition, and scene analysis
 - (11) Distributed artificial intelligence

“Artificial Intelligence” – Historical Definitions

John McCarthy, (1956.)

Marvin Minsky (MIT)

“AI is the science of making machines do things that require intelligence if done by men.”



“Artificial Intelligence” – Historical Definitions

Elain Rich (University of Texas at Austin)

“Artificial Intelligence is the study of how to make computers do things at which, at the moment, people are better.”



Eugene Charniak (Brown University)

“Artificial Intelligence is the study of mental faculties through the use of computational models.”



What is AI?

The science of making machines that:

Taxonomy of AI Definitions

- To reason **humanly**
 - *“The exciting new effort to make computers think .. machines with minds, in the full and literal sense”* (Haugeland, 1985)
 - *“The automation of activities that we associate with human thinking, activities such as decision-making, problem solving, learning .. ”* (Bellman, 1978)
- To reason **rationally**
 - *“The study of mental faculties through the use of computational models.”* (Charniak and McDermott, 1985)
 - *“The study of the computations that make it possible to perceive, reason, act.”* (Winston, 1992)

Taxonomy of AI Definitions

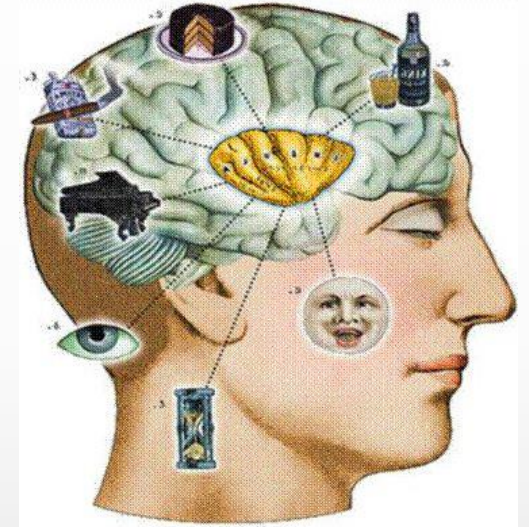
- To act **humanly**
 - *“The art of creating machines that perform functions that require intelligence when performed by people.”* (Kurzweil, 1990)
 - *“The study of how to make computers do things at which, at the moment, people do better.”* (Rich and Knight, 1991)
- To act **rationally**
 - *“The field of study that seeks to explain and emulate intelligent behavior in terms of computational processes.”* (Schalko, 1990)
 - *“The branch of computer science concerned with automation of intelligent behavior”* (Luger and Stubbleeld, 1993)

“Artificial Intelligence”

- “*Artificial*” means “man-made”
- “*Intelligence*” defines “*thinking power*”
- AI is “.. a branch of CS, by which we can create **intelligent machine**, which can behave like a **human**, think like a human, and able to **make decisions**”.
- The idea of AI is that you create a machine with programmed algorithm which can work with own intelligence (which is automatically present)
- What you do when working in AI? Design that program which has a man-made thinking power

AI-related disciplines

- **Humanities:** linguistics, philosophy, psychology
- **Natural sciences:** mathematics, biology
- **Cognitive science:** interdisciplinary scientific study of the mind (computing, neuroscience, psychology, linguistics, anthropology)

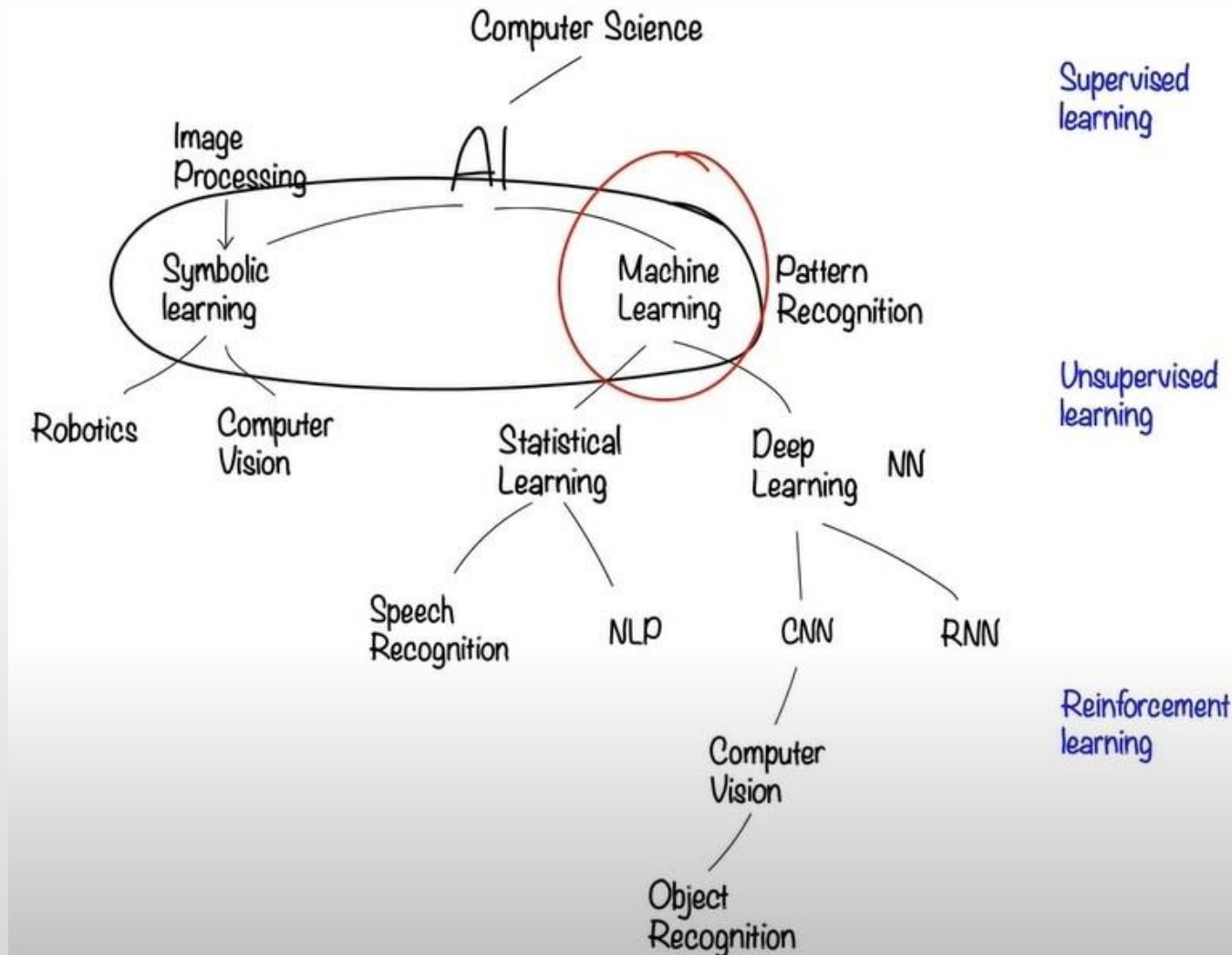


(Credit: J. Snajder, B Dalbelo-Basic)

Cognitive Science vs Artificial Intelligence

Cognitive science	Artificial intelligence
Intelligence	Artificial intelligence
Knowledge	Knowledge base
Cognition	Information processing
Learning	Machine learning
Learning/understanding language	Natural language processing

AI-related concepts



Classification

Prediction

Why AI?



*"AI can have **two** purposes.*

One is to use the power of computers to augment human thinking, just as we use motors to augment human or horsepower. Robotics and expert systems are major branches of that.

The other is to use a computer's artificial intelligence to understand how humans think. In a humanoid way. If you test your programs not merely by what they can accomplish, but how they accomplish it, then you're really doing cognitive science; you're using AI to understand the human mind."

- Herbert Simon